

	Work Instruction	Doc. Revision	01
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This procedure is applicable to the following models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD			
V810 / V810i XXL		√	√
V810 / V810i S2EX		√	√
V810 / V810i Mini			
V810 / V810i XLT / XLTM / XLL		√	√
V810 / V810i XLW			
V810 / V810i S3		√	√

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	15 Aug 2024	First Release	Yeap Leong Wei
01	30 Oct 2024	Include XXL for applicable model + content update	Yeap Leong Wei

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Scope

This procedure provides guidelines for upgrading the X-Axis Quantic Encoder Read Head and Scale.

Tool List

No	Part No	Description	Qty	UOM	Images
1	-	Ball Point Metric Allen Wrench Set	1	SET	
2	-	Metric T-Handle Wrench (M5 size)	1	PC	
3	-	Camera Tool	1	PC	
4	-	Temper Proof Screw Driver Set	1	Set	
5	-	Cutter	1	PC	

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6	4XASPP-A440	Renishaw Quantic Encoder Alignment Kit	1	Set	
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Table 1: Tool List

Upgrade Kit List

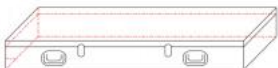
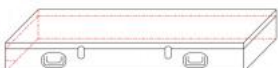
No	Part No	Description	Qty	UOM	Images
1	4XASPP-A424	RGH22 TO QUANTIC X-AXIS ENCODER KIT (S2EX)	1	SET	
2	4XASPP-A426	RGH22 TO QUANTIC X-AXIS ENCODER KIT (XLT)	1	SET	
3	4XASPP-A428	RGH22 TO QUANTIC X-AXIS ENCODER KIT (S3)	1	SET	
4	4FASCC-N385 -00	CABLE, X-AXIS ENCODER CONVERSION	1	PC	

Table 2: Upgrade Kit List

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Origin X-Axis Encoder Read Head Removal Procedure

1. **Board Unloading:** Prior to commencing the removal procedure, ensure that any boards within the machine are safely unloaded. Follow the manufacturer's guidelines or equipment manual for proper board removal procedures.
2. **Shutdown Sequence:** Close V810 GUI and initiate the shutdown sequence for the Super Micro PC by following the manufacturer's recommended procedure. Ensure all data is saved and any necessary backups are performed before proceeding.
3. **Deactivation of Equipment:** At the operator console panel, press the “X-Ray Off” button to deactivate any X-ray equipment. Turn the key to the “Off” position to ensure complete deactivation of the system and associated components.
4. **Power Down Procedure:** Safely power down the SSCA main power source by following the manufacturer's recommended shutdown procedure. Ensure all power indicators are off before proceeding with the removal process.
5. **Access Door Opening:** Open all front access doors carefully to provide sufficient access to the internal components of the machine. Ensure doors are fully opened and secured to prevent accidental closure during the removal process.
6. **Left Outer Barrier Removal:** Remove the left outer barrier by unscrewing the door screws using the appropriate tools. Store the screws in a secure location for reassembly after the removal process is complete.
 - a) For S2EX & XXL/XLT



Figure 1: Left Outer Barrier (S2EX & XXL/XLT)

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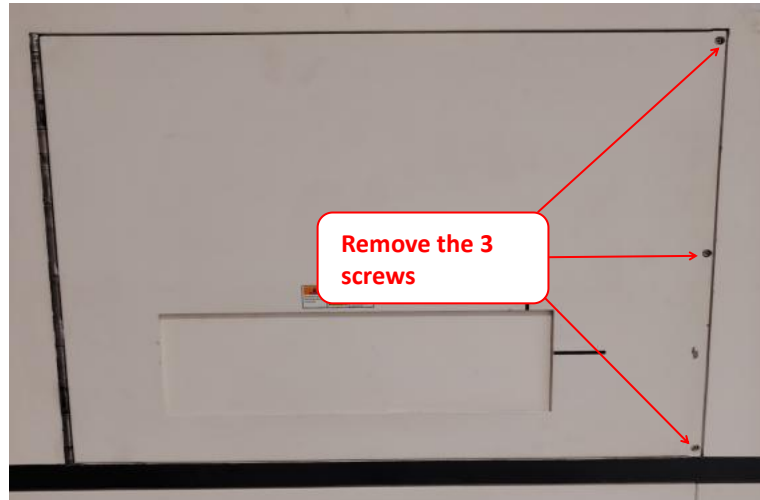


Figure 2: Left Outer Barrier (S3)

7. **Rear Access Door Opening:** Open the rear access door carefully, ensuring there are no cables or obstructions that may hinder the removal process. Take caution to prevent strain on any cables or connectors located near the access door.
8. **Removal of the vertical lead cover:** Unfasten the screws and remove the vertical lead cover. Take caution to prevent damage to the lead cover by placing it in flat and cleaned surface.

a) For S2EX, XXL/XLT

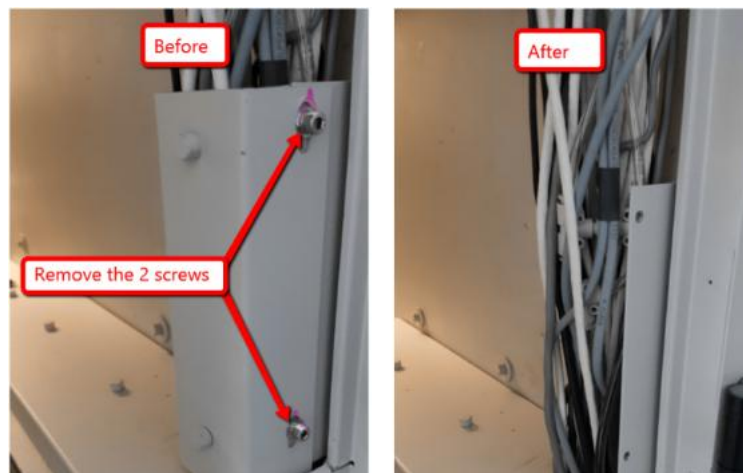


Figure 3: Vertical Lead Cover (Sample Image)

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b) For S3

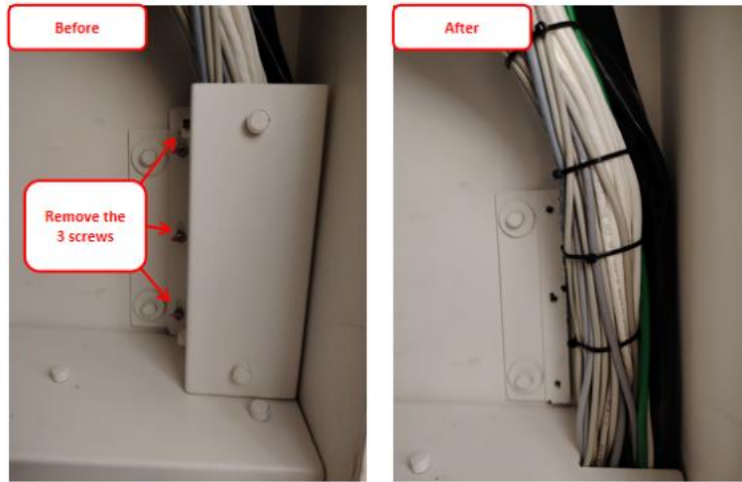


Figure 4: Vertical Lead Cover

9. **Removal of the horizontal lead cover:** Unfasten 3 screws and remove the horizontal lead cover. Take caution to prevent damage to the lead cover by placing it in flat and cleaned surface.

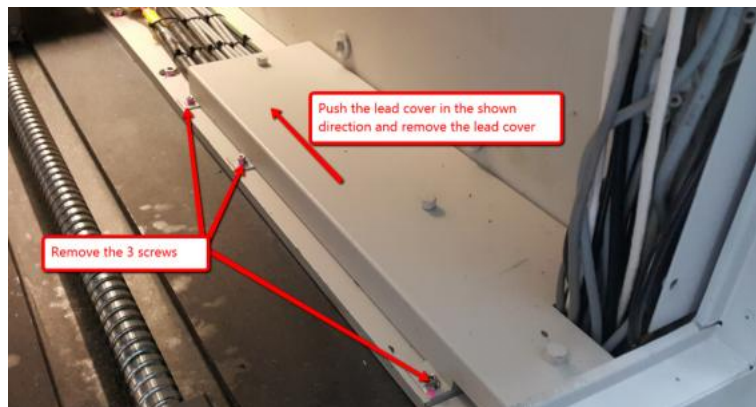


Figure 5: Horizontal Lead Cover (Sample Image)

10. **Removal of the SSCA Cable Bracket or Cable Tray Cover:** Remove the bracket or cable tray cover for the facilitation of the cable removal process.

a) For S2EX and XXL/XLT

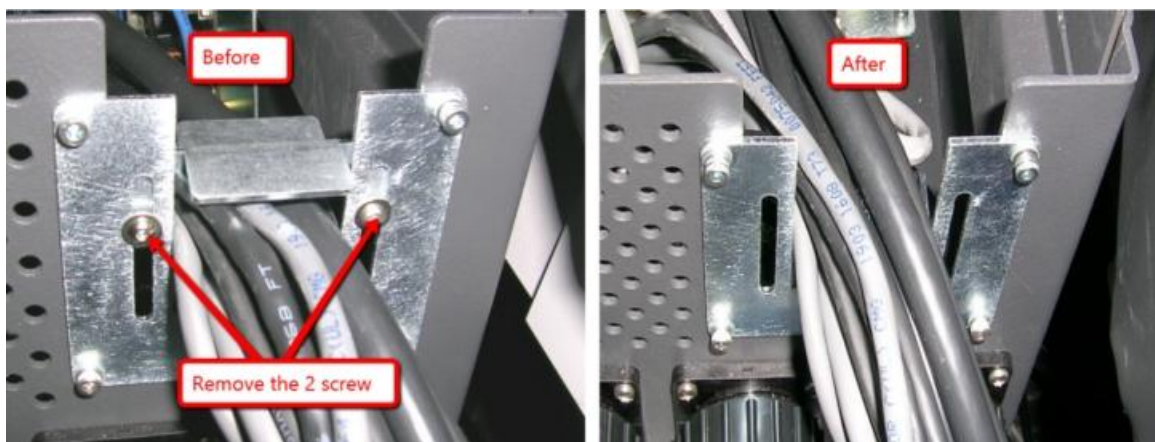


Figure 6: SSCA (Sample Image)

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b) For S3



Figure 7:SSCA for S3 (Sample Image)

11. **Disconnected the X-Axis Encoder Head Connector:** Unplug X-Axis read head encoder connector from X-Axis Kollmorgen kit (for S3), from conversion cable (for S2EX) or from the extension cable connector (for XXL / XLT).

Note: Remove necessary cable ties and carefully remove the encoder cable.

a) For XXL / XLT

Note: For XXL / XLT, please unplug encoder read head from the extension cable connector.



Figure 8:X-Axis Kollmorgen Kit

b) For S3

Note: Please unplug the encoder read head from the kollmorgen kit.

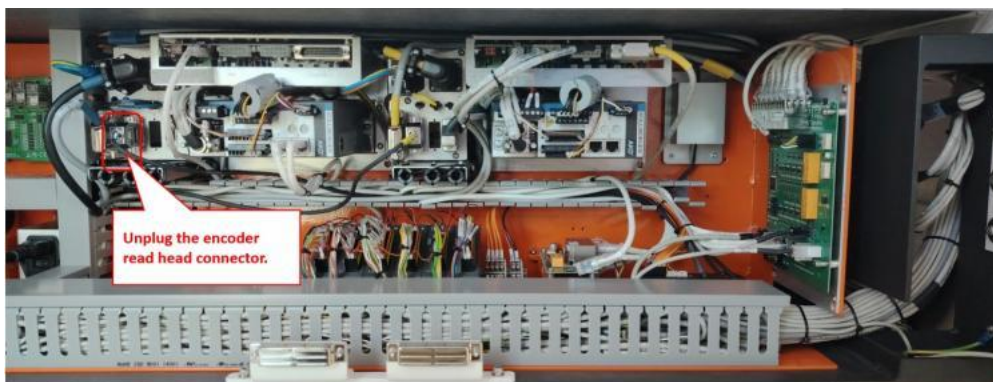


Figure 9:X-Axis Kollmorgen Kit

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12. **Removal of X-Axis Read Head Encoder:** Remove X-Axis Read Head Encoder by referring to the routing below.

Note: Follow the red arrow for removal direction, and remove any necessary cable ties.

a) For S2EX (Part ID: 4JASC-A082) and XXL / XLT (Part ID: 4XASP-A095)

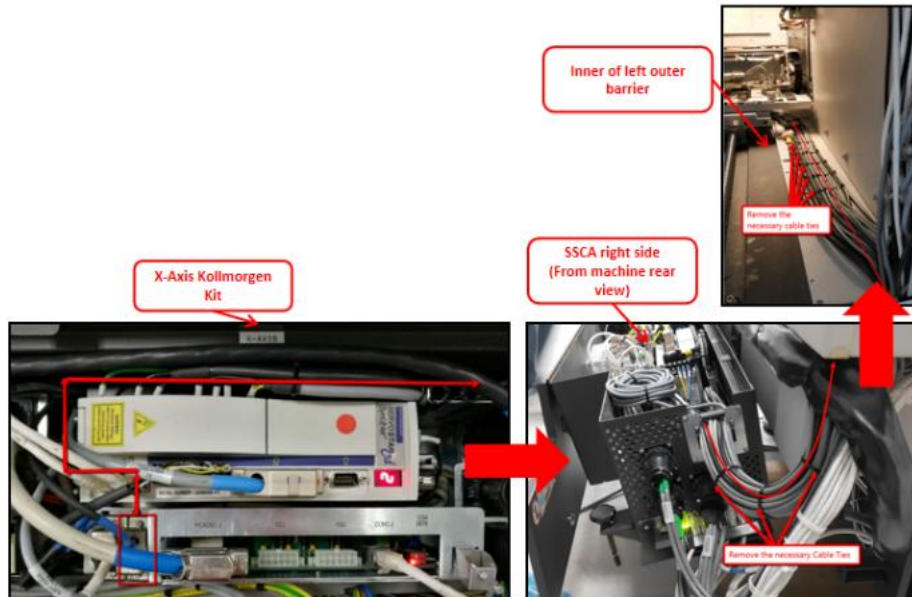


Figure 10:X-Axis Read Head Encoder cable routing (Sample Image)

b) For S3(Part ID: 4FASCC-A082)

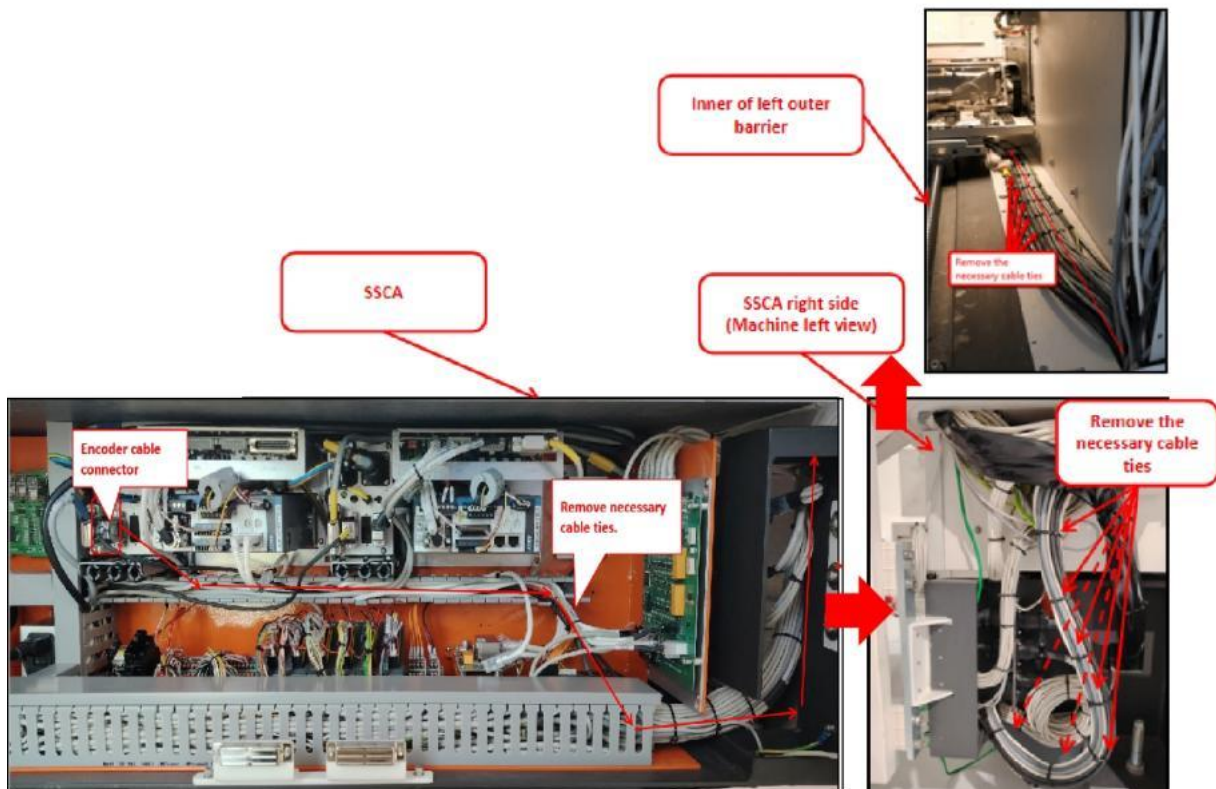


Figure 11:X-Axis Read Head Encoder cable routing

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13. **Cable Tie Removal:** Back to machine front side, remove any cable ties from the cable tray that may be securing the encoder cables. Use cable tie removal tools if necessary to avoid damage to the cables or surrounding components.

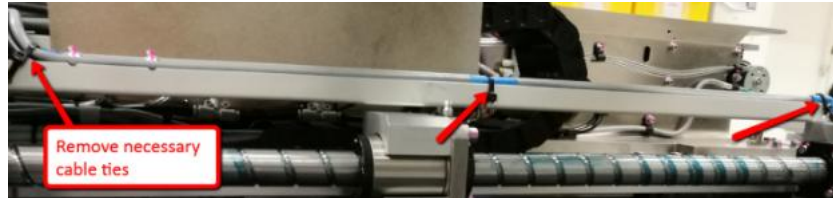


Figure 12: X-Axis Read Head Encoder cable routing

14. **X-Axis Motor Cable Disconnection:** Disconnect the X-Axis motor cable connector from its corresponding port. Remove any screws securing the cable to the cable tray and set them aside for reassembly.

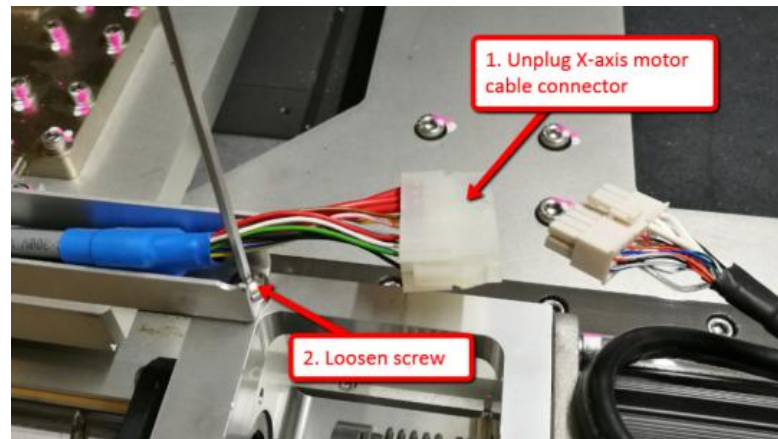


Figure 13: X-Axis Rotary Motor (Sample Image)

15. **X-Axis E-Chain Guard Plate and Bracket Removal:** Unfasten the screws securing the X-Axis E-chain guard plate and motor cable bracket. Carefully remove these components and store them in a safe location for re-installation.

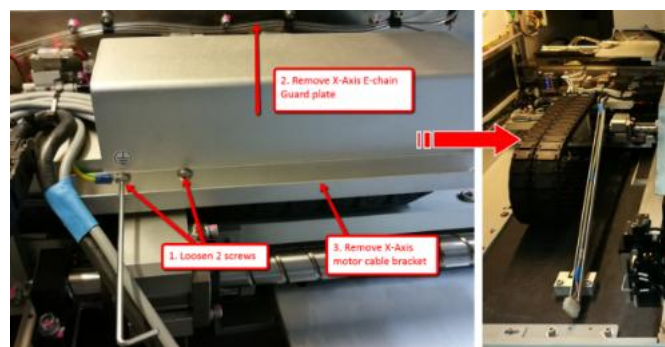


Figure 14: X-Axis Cable Tray and E-Chain Guard Plate (Sample Image)

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16. **Cable Clamp Removal:** Unfasten the screws securing the cable clamp and remove it from the assembly. Take care not to damage the cables or surrounding components during removal.



Figure 15:X-Axis Read Head Encoder (Sample Image)

17. **X-Axis Read Head Encoder Bracket Removal:** Using a camera tool for precision, unfasten the screws securing the X-Axis read head encoder bracket. Carefully remove the encoder, ensuring gentle handling to prevent any damage.

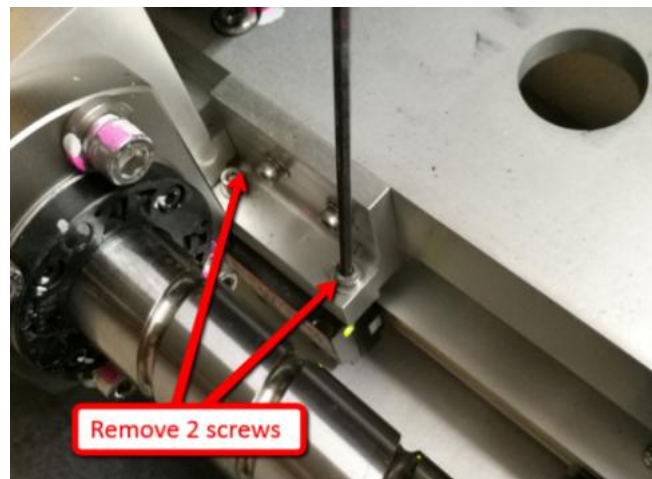


Figure 16:X-Axis Read Head Encoder Bracket (Sample Image)

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18. **X-Axis E-Chain Slot Cover Opening:** Use a flat-head screwdriver to carefully open the first X-Axis E-chain slot cover. Apply gentle pressure to avoid damaging the cover or surrounding components.

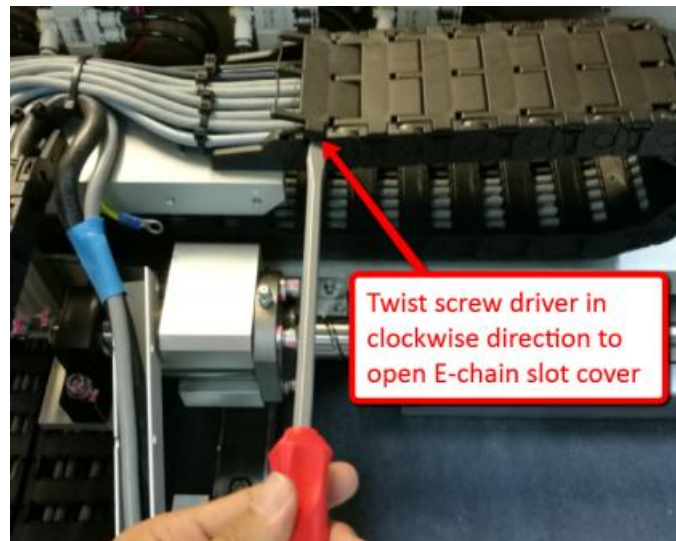


Figure 17: X-Axis E-Chain

19. **Cable Tie Removal:** Remove any necessary cable ties from the cable tray, ensuring that cables are free and unobstructed for removal. Take care to avoid damaging the cables or surrounding components.

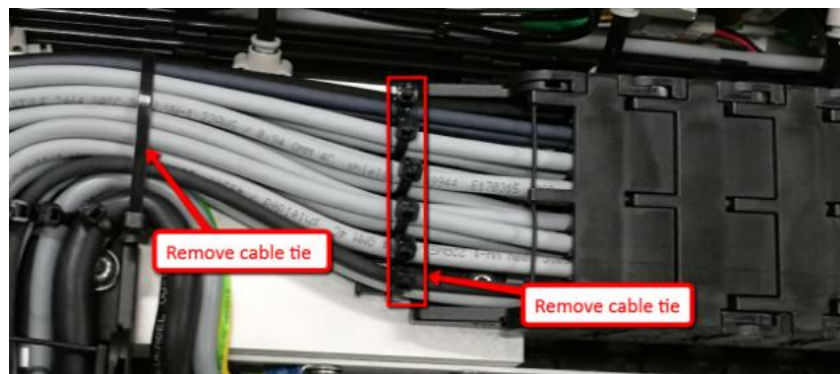


Figure 18: X-Axis E-Chain (Partial cover removed)

20. **X-Axis E-Chain Top Joint Opening:** Use a flat-head screwdriver to carefully open the X-Axis E-chain top joint. Ensure smooth movement and avoid any strain on the cables or connectors.

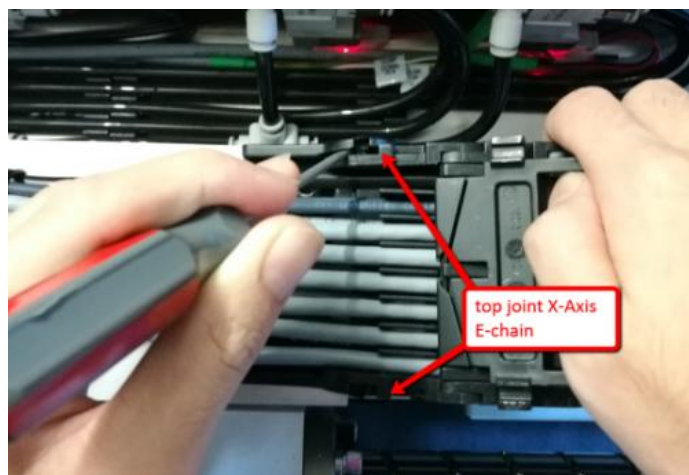


Figure 19: X-Axis E-Chain (Top Joint)

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21. **Linear Encoder Cable Removal:** Loosen the screws securing the linear encoder cable and carefully remove it from the assembly. Pay attention to the cable routing to ensure smooth removal without causing any damage.

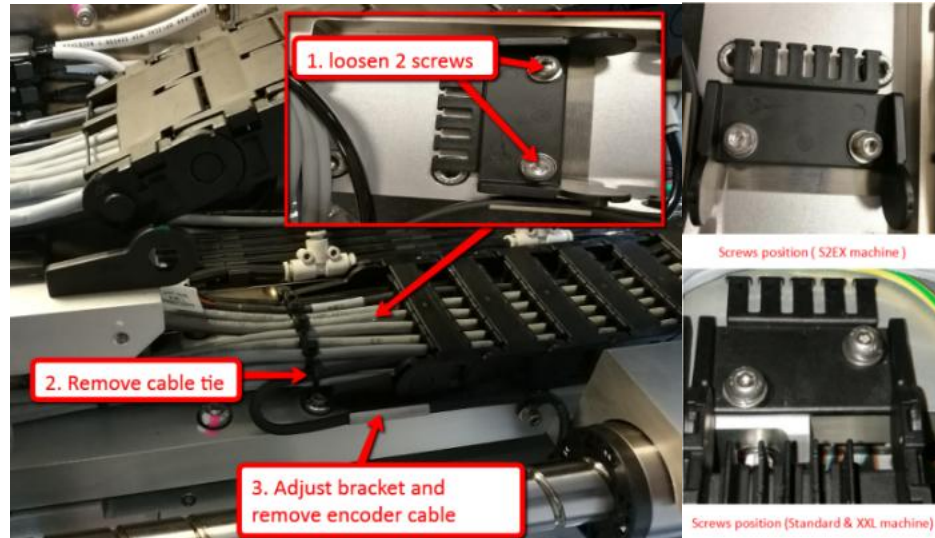


Figure 20: X-Axis E-Chain

22. **X-Axis E-Chain Bottom Joint Opening:** Use a flat screwdriver to gently open the X-Axis E-chain bottom joint. Take care not to exert excessive force to prevent damage to the cables or connectors.

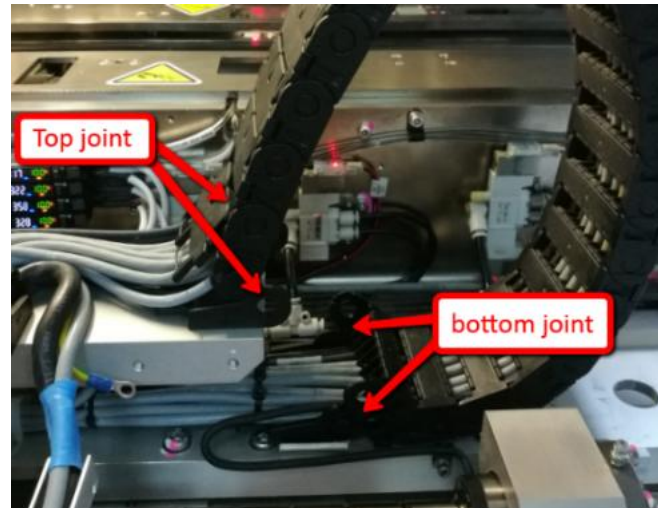


Figure 21: X-Axis E-Chain (Top & Bottom Joint)

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23. **Remaining X-Axis E-Chain Slot Cover Opening:** Open the remaining X-Axis E-chain slot covers systematically, applying cable ties after opening every 2-3 slot covers to maintain organization and prevent tangling.

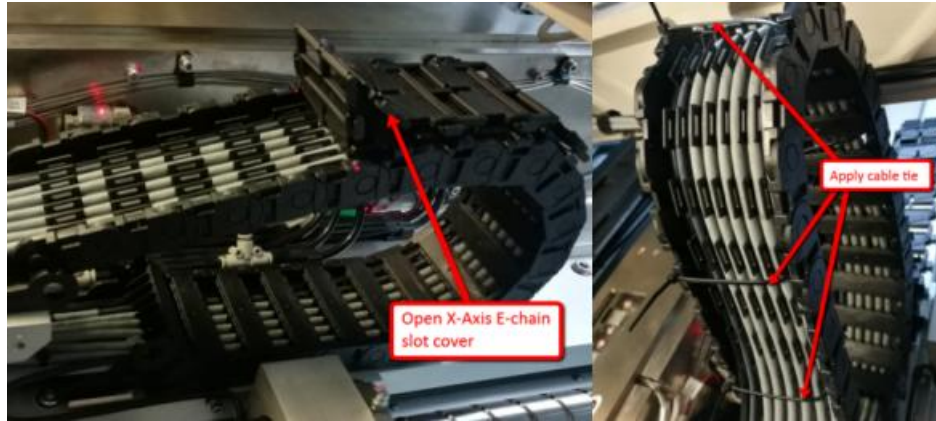


Figure 22: X-Axis E-Chain

24. **X-Axis Read Head Encoder Cable Removal:** Gently remove the X-Axis read head encoder cable from the X-Axis E-chain, ensuring smooth movement and avoiding any damage to the cables or connectors.
25. **Additional Cable Tie Removal and Y-Axis E-Chain Slot Cover Opening:** Remove any necessary cable ties, using a flat-head screwdriver to open end joints and slot covers on the Y-Axis wide E-chain (right-hand side). Push the stage to the rear side if needed for easier access.

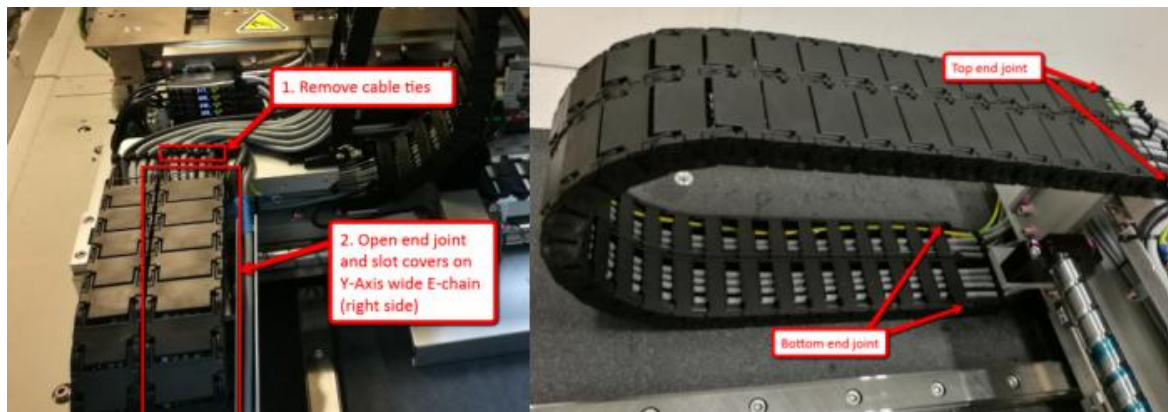


Figure 23: Y-Axis E-Chain

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26. **X-Axis Linear Encoder Cable Removal:** Gently remove the X-Axis linear encoder cable from the X-Axis E-chain, ensuring smooth movement and avoiding any damage to the cables or connectors.
27. **Remaining Y-Axis E-Chain Slot Cover Opening and Cable Tie Application:** Open the remaining Y-Axis wide E-chain slot covers systematically, applying cable ties only to non-related cables to maintain organization and prevent tangling.
28. **X-Axis Encoder Cable Removal from Y-Axis E-Chain:** Gently remove the X-Axis encoder cable from the Y-Axis E-chain, ensuring that it is free from any entanglement or obstruction.
29. **Final X-Axis Encoder Removal:** Gently remove the X-Axis encoder from the machine, ensuring that it is handled carefully to prevent any damage during removal and transport.
 - a) S2EX machine (Part ID: 4JASC-A082)
 - b) XXL / XLT machine (Part ID: 4XASP-A095)
 - c) S3 machine (Part ID: 4FASCC-A082)

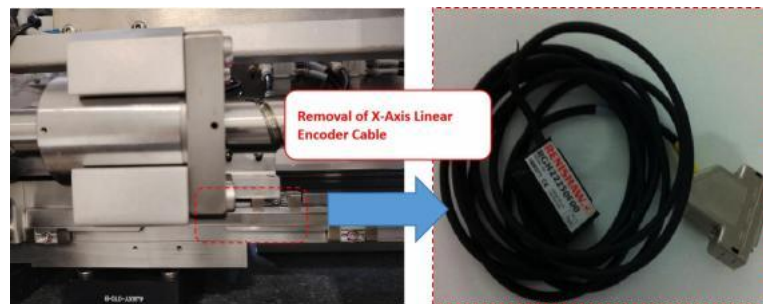


Figure 24: Removal of X-Axis Encoder Head Cable

30. **Proceed for glass scale removal and upgrade to Quantic Scale:** Refer to procedure **Origin X-axis Glass Scale Removal Procedure**.

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Origin X-Axis Glass Scale Removal Procedure

31. **Dismantle the home sensor and limit sensor:** Unscrew the M3 screw for home sensor and limit sensor and remove it from the machine.
 - a) S2EX machine (Part ID: 4XAXY-055 & 4XAXY-057)
 - b) XXL / XLT machine (Part ID: 4XAXY-055 & 4XAXY-057)
 - c) S3 machine (Part ID: 4FAHWP-A155 & 4FAXYS-A108)
32. **Loosen the Screws for Spring Clamp:** Loosen the screws for the spring clamp with M3 Allen key.

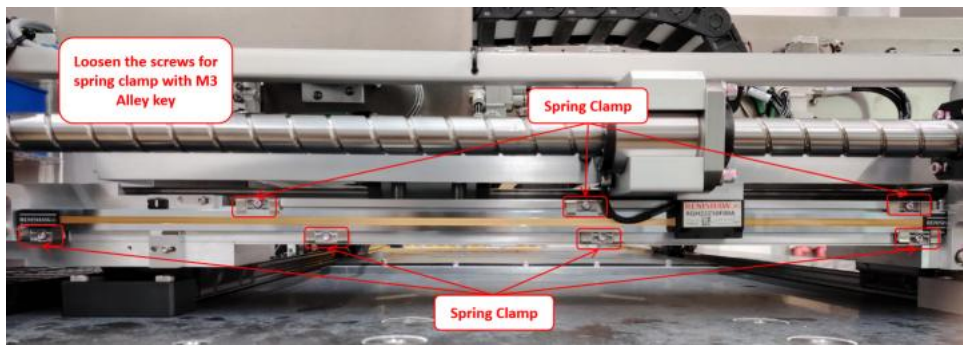


Figure 25: X-Axis Glass Scale

33. **Removal of Steel Bar with Clamps (Top Side):** Unfasten the screws for steel bar with clamps. Carefully remove and place it on the clean and flat surface.



Figure 26: X-Axis Cover Plates

34. **Removal of X-Axis Quantic Scale:** Carefully remove the X-Axis Glass scale. Take caution to place the scale in a flat surface to prevent any incident.



Figure 27: Glass Scale

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New X-Axis Quantic Scale Replacement

35. **Spare Part Preparation:** Refer to the table below to gather and prepare the new spare part required for the replacement process.

a) For S2EX

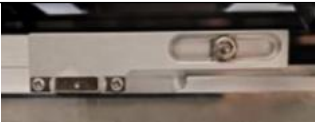
Part number	Part Description	QTY	UOM	illustration
4XASPP-A430	S2EX X-AXIS QUANTIC ENCODER SCALE ASSEMBLY KIT	1	PC	
4JAXYS-A602	S2EX HOME SWITCH ASSEMBLY, X-AXIS	1	PC	
4JAXYS-A604	S2EX LIMIT SWITCH ASSEMBLY, X-AXIS	1	PC	

Table 3: X-Axis Quantic Spare Part List for S2EX

b) For XXL / XLT

Part number	Part Description	QTY	UOM	illustration
4XASPP-A432	XLT X-AXIS QUANTIC ENCODER SCALE ASSEMBLY KIT	1	PC	
4CAXYS-A602	XLT HOME SWITCH ASSEMBLY, X-AXIS	1	PC	
4CAXYS-A604	XLT LIMIT SWITCH ASSEMBLY, X-AXIS	1	PC	

Table 4: X-Axis Quantic Spare Part List for XXL / XLT

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c) For S3

Part number	Part Description	QTY	UOM	illustration
4XASPP-A434	S3 X-AXIS QUANTIC ENCODER SCALE ASSEMBLY KIT	1	PC	
4FAXYS-A602	S3 HOME SWITCH ASSEMBLY, X-AXIS	1	PC	
4FAXYS-A604	S3 LIMIT SWITCH ASSEMBLY, X-AXIS	1	PC	
4FAXYS-N274-00	S3 , X-AXIS ENCODER CLAMP MOUNT BAR	1	PC	

Table 5: X-Axis Quantic Spare Part List for S3

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36. **X-Axis Quantic Scale Assembly:** Assemble the Quantic scale and allow it to rest on the spring clamps. Be cautions for any obstacle and not damaging the Quantic scale.

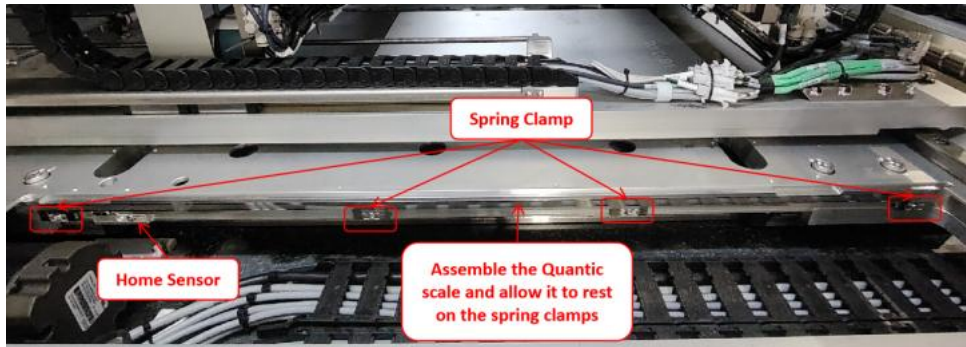


Figure 28:X-Axis Quantic Scale (Sample Image)

37. **Steel Bar with Clamps (Top side) Assembly:** Assemble the steel bar with clamps that guiding the top side of the Quantic scale. Tighten the M3 screws after align the steel bar in position. For S3 machine, it will have to be replace with new part as per following
- a) For S2EX and XXL / XLT, use back the original clamp mount bar
 - b) For S3 machine (Part ID: 4FAXYS-N274-00)

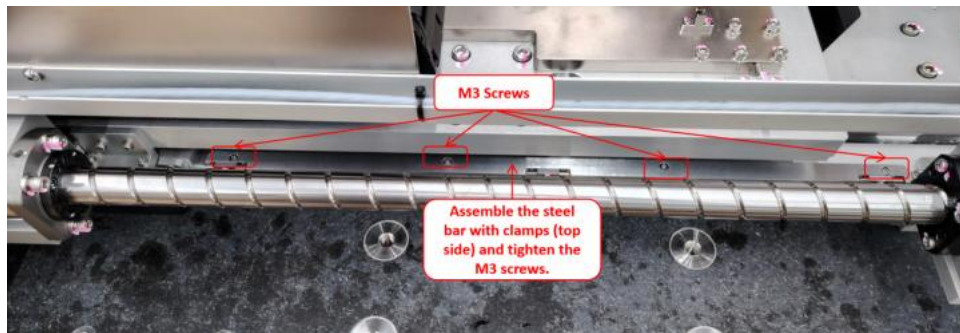


Figure 29:X-Axis Encoder Cover Plate and Limit Sensor (Sample Image)

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38. **Home and Limit Sensors Assembly:** Align the home sensor and tighten the screw in the center position of the elongate. Then, assemble the limit sensor and tighten the M3 screw.

- S2EX machine (Part ID: 4JAXYS-A602 & 4JAXYS-A604)
- XXL / XLT machine (Part ID: 4CAXYS-A602 & 4CAXYS-A604)
- S3 machine (Part ID: 4FAXYS-A602 & 4FAXYS-A604)

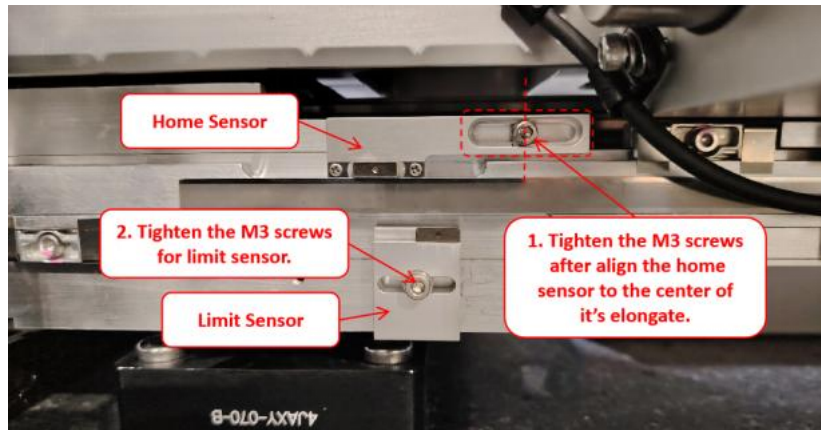


Figure 30: X-Axis Home and Limit Sensors Alignment

39. **Replace the New Quantic Encoder:** Proceed to replace the X-Axis Quantic Encoder Read Head follow the procedure “**New X-Axis Encoder Read Head Replacement Procedure**”.

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New X-Axis Encoder Read Head Replacement Procedure

40. **Spare Part Preparation:** Refer to the table below to gather and prepare the new spare part required for the replacement process.

a) For S2EX

Part number	Part Description	QTY	UOM	Illustration
2675-0053	Encoder Head, Optical, 0.5UM	1	PC	
2720-0069	SOCKET HEAD SCREW,HEX,S/S,M2.5,10	2	PC	
4JAXYS-N520-00	X-AXIS ENCODER READHEAD ADAPTER	1	PC	
4JASCC-A259	S2EX, LINEAR ENCODER X-AXIS EXTENSION	1	PC	

Table 6: Spare Part List for S2EX

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b) For XXL / XLT

Part number	Part Description	QTY	UOM	Illustration
2675-0053	Encoder Head, Optical, 0.5UM	1	PC	
2720-0069	SOCKET HEAD SCREW,HEX,S/S,M2.5,10	2	PC	
4JAXYS-N520-00	X-AXIS ENCODER READHEAD ADAPTER	1	PC	
4CASCC-A259	XLT, LINEAR ENCODER X-AXIS EXTENSION	1	PC	

Table 7: Spare Part List for XXL / XLT

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c) For S3

Part number	Part Description	QTY	UOM	Illustration
2675-0053	Encoder Head, Optical, 0.5UM	1	PC	
2720-0069	SOCKET HEAD SCREW,HEX,S/S,M2.5,10	2	PC	
4JAXYS-N520-00	X-AXIS ENCODER READHEAD ADAPTER	1	PC	
4JASCC-A259	S2EX, LINEAR ENCODER X-AXIS EXTENSION	1	PC	
4FASCC-N385-00 (Optional: Only applicable to S3 w/ AKD or S200 driver)	CABLE, X-AXIS ENCODER CONVERSION	1	PC	

Table 8: Spare Part List for S3

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41. **Encoder Assembly:** Assemble the X-Axis Encoder to the read head adapter.

- a) S2EX machine (Part ID: 2675-0053*1, 4JAXYS-N520-00*1, 2720-0069*2)
- b) XXL / XLT machine (Part ID: 2675-0053*1, 4JAXYS-N520-00*1, 2720-0069*2)
- c) S3 machine (Part ID: 2675-0053*1, 4JAXYS-N520-00*1, 2720-0069*2)

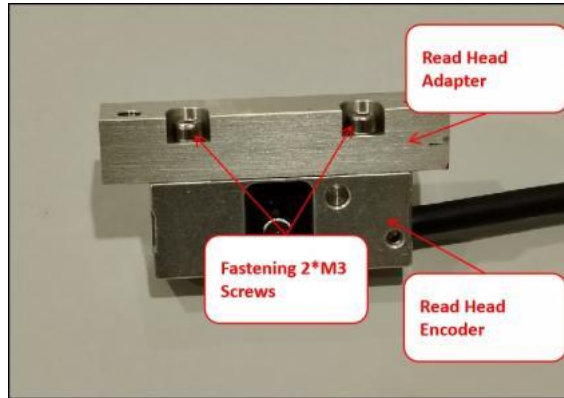


Figure 31: Securing X-Axis Read Head Encoder to Adapter

42. **Encoder Bracket Assembly:** Securely tighten the two screws to ensure proper attachment.

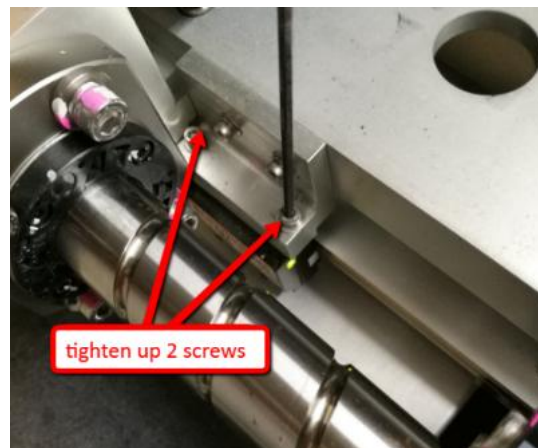


Figure 32: Attaching X-Axis Read Head Encoder Bracket

43. **Cable Clamp Application:** Apply the cable clamp to secure the encoder cables in place and tighten the two screws firmly.



Figure 33: X-Axis Ball Screw Bearing (Sample Image)

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44. **Read Head Encoder Cable Installation:** Securely clamp back the X-Axis read head encoder cable and tighten the two screws to ensure stable connection.

Note: Ensure the cable holder is full press inside before securing the screws.

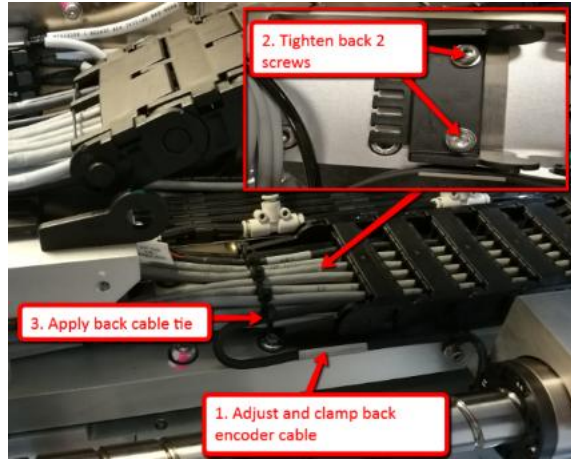


Figure 34:X-Axis E-Chain (Sample Image)

45. **Cable Routing and E-Chain Closure:** Gently route the X-Axis encoder cable back to its original position, removing any necessary cable ties for flexibility. Close the E-chain slot cover to each partition securely.

Note: If machine using the Tribo Tape mode, perform cable management nicely to avoid cable twisted.



Figure 35:X-Axis E-Chain (Sample Image)

46. **E-Chain End Joint Closure and Cable Management:** Close the E-chain end joints on both sides and apply necessary cable ties to maintain proper cable organization and routing.

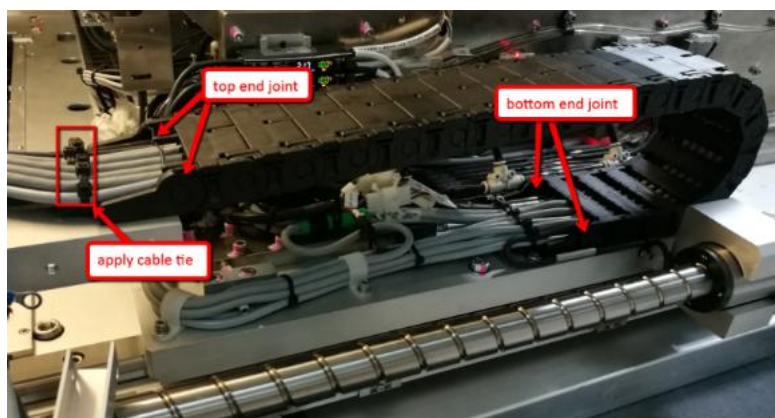


Figure 36:X-Axis E-Chain End Joint (Top & Bottom)

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47. **Stage Movement and Cable Inspection:** Push the X-Axis stage to the left and right several times to ensure smooth movement. Verify that cables are well-routed within the E-chain. If any issues arise, open the E-chain and re-route the cables as needed.

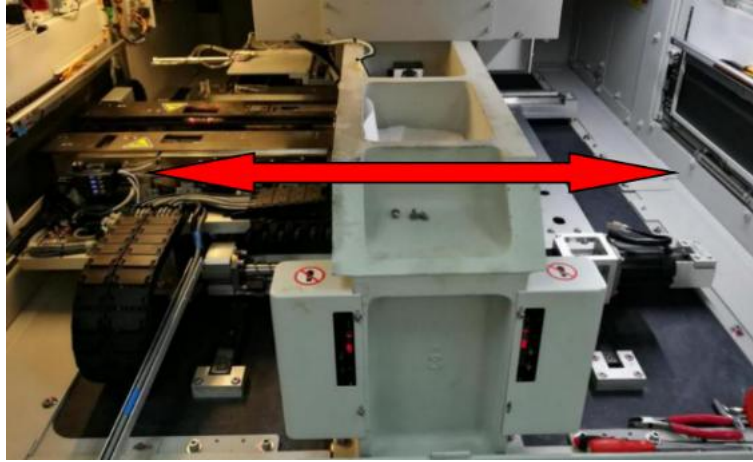


Figure 37: X-Axis Stage (Sample Image)

48. **Bracket and Guard Plate Assembly:** Assemble the X-Axis motor cable bracket and guard plate, securely tightening the two screws. Note that the grounding wire is screwed together with the guard plate for proper grounding.

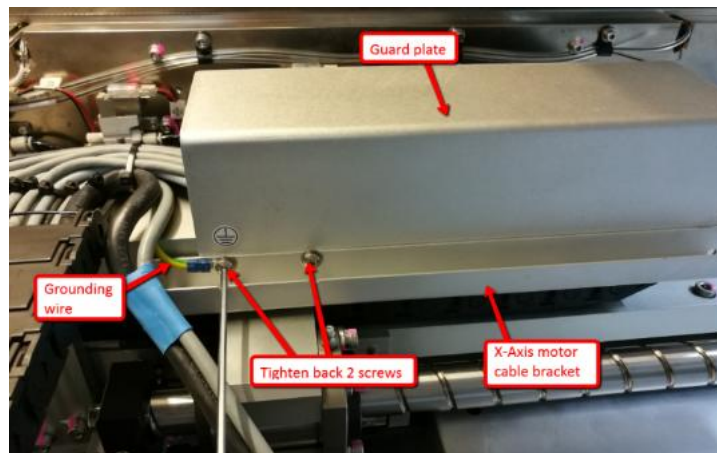


Figure 38: X-Axis Cable Tray and Guard Plate

49. **Motor Cable Connector Re-connection:** Tighten the screw and plug back the X-Axis motor cable connector to restore electrical connection.

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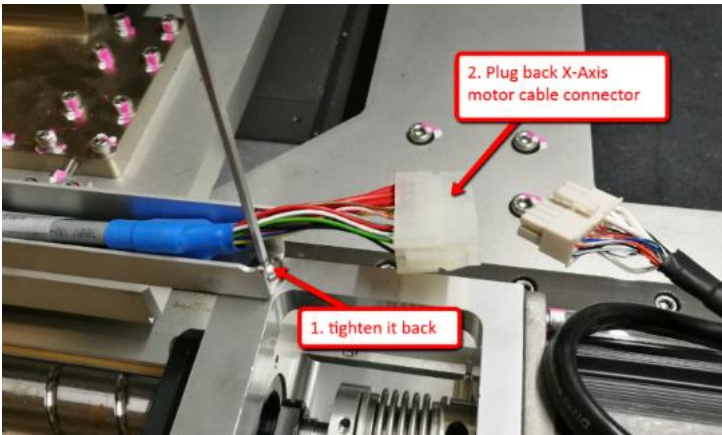


Figure 39:X-Axis Rotary Motor Housing

50. **Encoder Cable Routing to Y-Axis E-Chain:** Gently route the X-Axis encoder cable to the Y-Axis E-chain. Close the E-chain slot covers to each partition according to the provided diagram. Ensure all cables are not twisted to prevent interference.

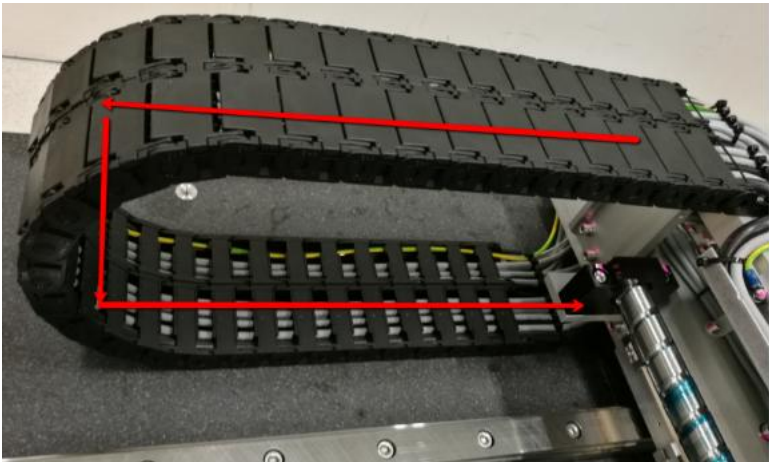


Figure 40:Y-Axis E-Chain

51. **Y-Axis E-Chain Closure and Cable Management:** Close the end joints on both sides of the Y-Axis E-chain and apply necessary cable ties to maintain proper cable organization and routing.

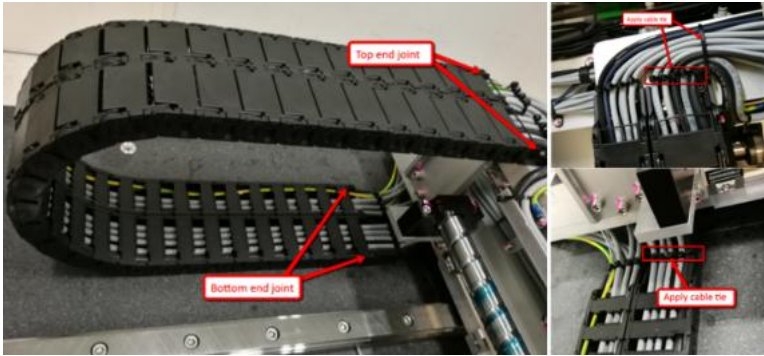


Figure 41:Y-Axis E-Chain End Joints (Top & Bottom)

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52. **Stage Movement and Cable Inspection (Y-Axis):** Move the Y-Axis stage to the front and rear several times to ensure smooth movement. Verify that cables are well-routed within the E-chain. If any issues arise, open the E-chain and re-route the cables as needed.

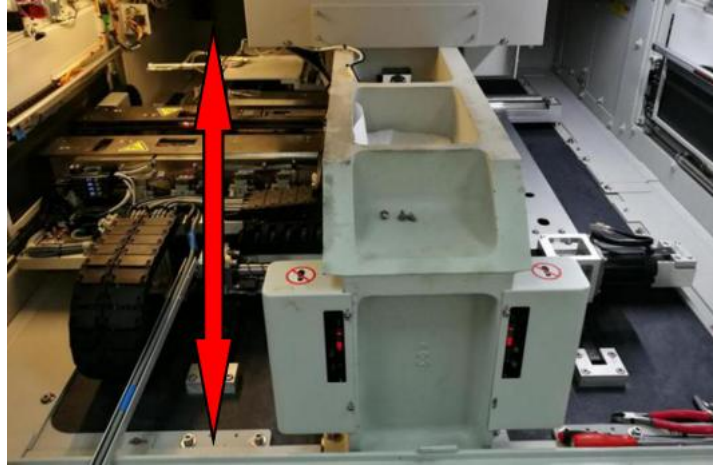


Figure 42: Y-Axis Stage

53. **Read Head Encoder Connection with Extension Cable:** Connect the read head encoder with the extension cable interface to the SSCA to restore full functionality. Ensure proper wiring management for the extension cable. Refer extension cable part ID per following.
- S2EX machine (Part ID: 4JASCC-A259)
 - XXL / XLT machine (Part ID: 4CASCC-A259)
 - S3 machine (Part ID: 4JASCC-A259)

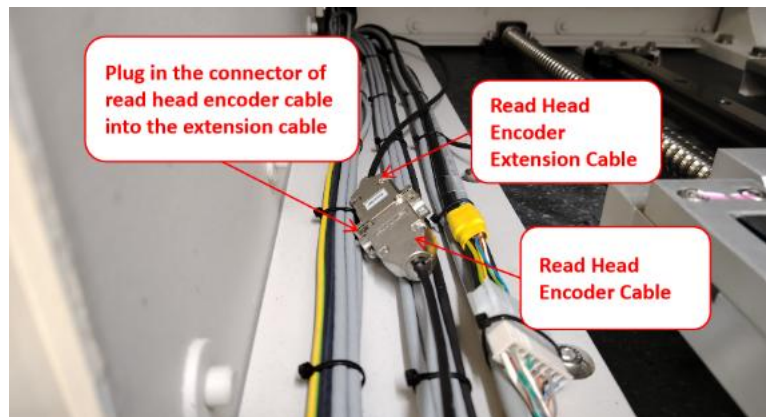


Figure 43: X-Axis Encoder Cable Connection

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54. **Connect the X-Axis Extension Cable Connector:** Plug in the X-Axis Extension Cable connector to the conversion cable interfaced to X-Axis Kollmorgen kit.

Note: S3 with Panasonic driver will direct connect to kollmorgen kit.

a) For S2EX and XXL / XLT



Figure 44: X-Axis Kollmorgen Kit

b) For S3 (Part ID: 4FASCC-N385-00, only applicable to AKD/S200 driver)

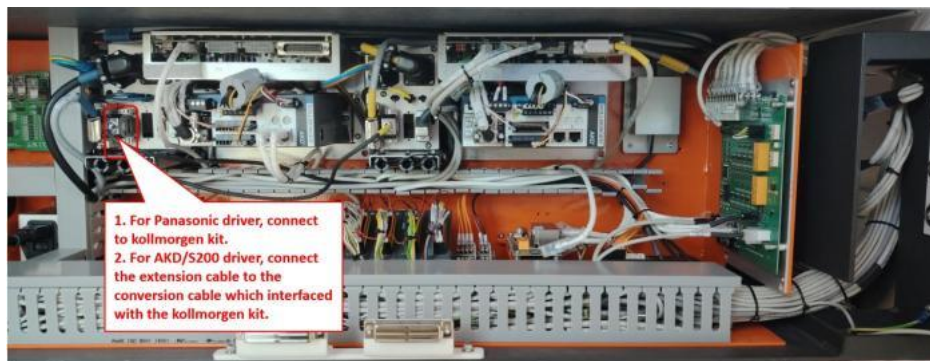


Figure 45: X-Axis Kollmorgen Kit

55. **Cable Clamp/Tray Assembly:** Place the cable clamp on its position and tighten back the 2 screws. For S3, ensure to close the cable tray.

a) For S2EX and XXL / XLT

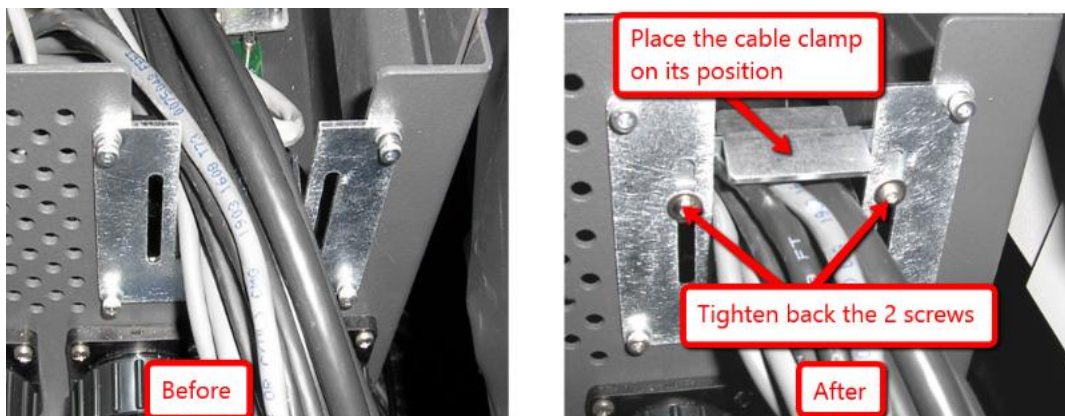


Figure 46: SSCA (Sample Image)

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b) For S3



Figure 47:SSCA

56. **Horizontal Lead Cover Assembly:** Place back the horizontal lead cover and push it in the direction as shown in below figure. Then, tighten back the 3*M5 screws.

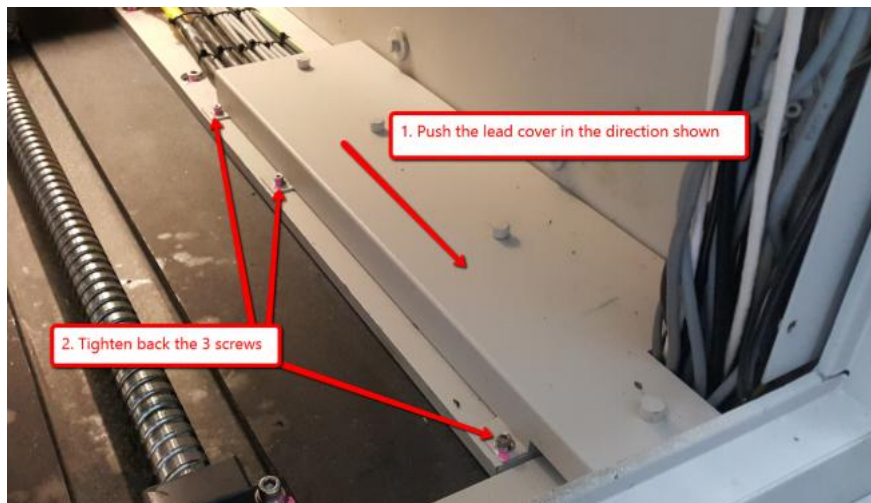


Figure 48:Horizontal Lead Cover (Sample Image)

57. **Vertical Lead Cover Assembly:** Assemble the vertical lead cover and tighten the screws.

a) For S2EX and XXL / XLT



Figure 49:Vertical Lead Cover (Sample Image)

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b) For S3

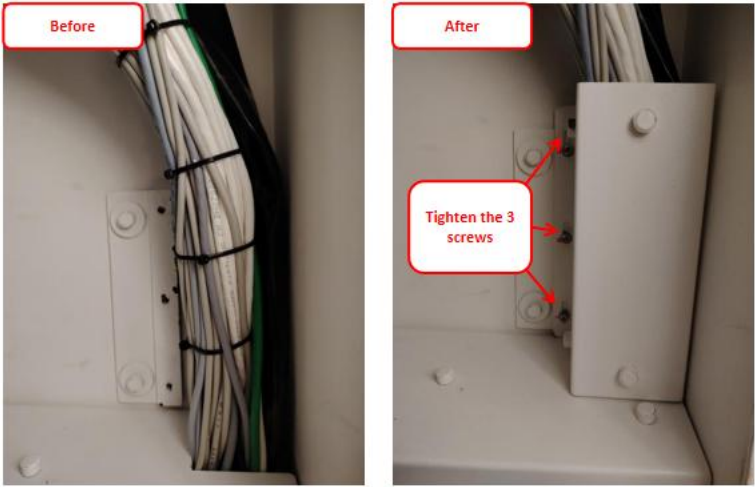


Figure 50: Vertical Lead Cover

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Tool Removal and Cleaning

58. Before concluding the replacement procedure, carefully inspect the machine interior to ensure that all tools, equipment, and debris have been removed. Retrieve any tools or equipment left inside the machine and return them to their designated storage area to prevent any interference with machine operation.
59. Dispose of any removed cable ties, packaging materials, or other debris in the designated waste disposal containers according to company guidelines or environmental regulations.
60. Once the interior surfaces have been cleaned and cleared of debris, visually inspect the machine to ensure that it is in a tidy and organized state, ready for operation.
61. Perform a final check to verify that all components have been properly installed and secured according to manufacturer specifications. Confirm that all cables are properly routed and secured within their respective E-chains to prevent any interference with machine movement.
62. Once the machine has been cleaned and inspected, power on the system according to standard operating procedures but without launch the V810 GUI.

Reference for Alignment Procedure

63. **X-Axis Encoder Read Head Alignment:** Refer to “AXI V810 Quantic Encoder Read Head Alignment Procedure” to verify the encoder status and perform necessary alignment if needed.
Note: Crucial to ensure the signal strength for encoder read head along the x-axis stroke.
64. **Home and Limit Sensors Alignment:** After validating the encoder head performance, refer to **AXI V810 Quantic Home and Limit Sensors Upgrade Alignment Procedure (AKD, S200, Panasonic)** and perform necessary alignment and adjustment.
Note: It's a must perform procedure whenever there is any adjustment or replacement of Quantic scale.